SAVES v5.0

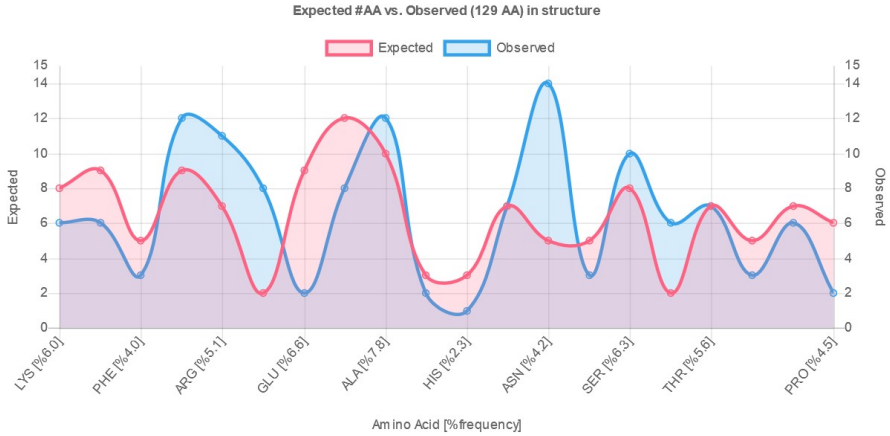
WHATCHECK • PROCHECK • ERRAT • Verify3D • PROVE • CRYST • pdbU ?

New Job

VERIFY	ERRAT	PROVE	WHATCHECK
100.00% of the residues have averaged 3D-1D score >= 0.2 **Pass**	Overall Quality Factor **A: 97.5207**	Buried outlier protein atoms total from 1 Model: 3.6% Warning	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47

Job results for:model.pdb | Link to this job:600724
Interactive Ramachandran Plot | View Structure

finished: Nov 2nd, 2020 [6:17 AM]
Amino acid distributions determined from 168,799 PDB structures (2020/10/28)



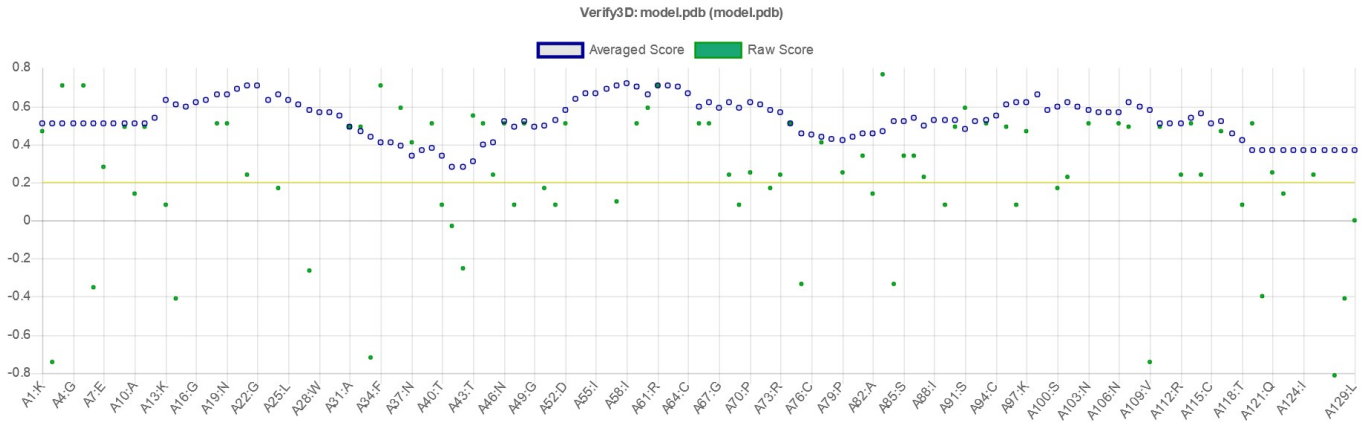
Verify 3D results

↑ TOP

100.00% of the residues have averaged 3D-1D score >= 0.2

Pass

At least 80% of the amino acids have scored >= 0.2 in the 3D/1D profile.



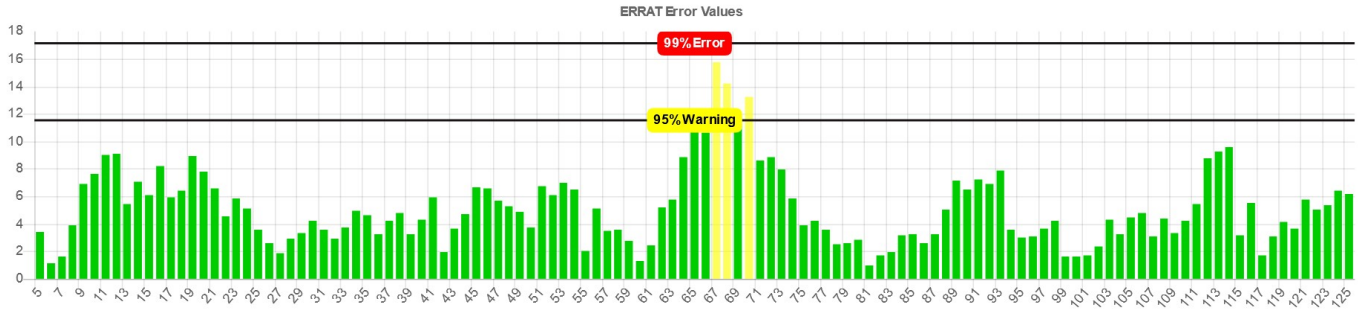
Save as image Save data to file

ERRAT results

↑ TOP

Input: model.pdb (model.pdb)
Moleman is used to identify chains and separate into files. Each pdb chain file is linked below for each plot. For an explanation on how the chains were found, here is the moleman logfile

Quality Factor: A: 97.5207 | PDF | PostScript | Log | PDB chain file used



Save as image

PROVE results

↑ TOP

Input file: **model.pdb**
Model : 17 buried outlier protein atoms, 3.6% Warning
Job Run Report | Labeled pdb

- **Job output**
- **Plot PS** | **Plot PDF**
- **Job run log**

WHATCHECK results

↑ TOP

1. Error: Missing unit cell information **Error** ↑ TOP

```
No SCALE matrix is given in the PDB file.
```

2. Error: Missing symmetry information **Error** ↑ TOP

```
Problem: No CRYST1 card is given in the PDB file.
```

3. Note: No rounded coordinates detected Pass ↑ TOP

```
No significant rounding of atom coordinates has been detected.
```

4. Note: Valine nomenclature OK Pass ↑ TOP

```
No errors were detected in valine nomenclature.
```

5. Note: Threonine nomenclature OK Pass ↑ TOP

```
No errors were detected in threonine nomenclature.
```

6. Note: Isoleucine nomenclature OK Pass ↑ TOP

```
No errors were detected in isoleucine nomenclature.
```

7. Note: Leucine nomenclature OK Pass ↑ TOP

```
No errors were detected in leucine nomenclature.
```

8. Note: Arginine nomenclature OK Pass ↑ TOP

```
No errors were detected in arginine nomenclature.
```

9. Note: Tyrosine torsion conventions OK Pass

↑ TOP

No errors were detected in tyrosine torsion angle conventions.

10. Note: Phenylalanine torsion conventions OK Pass

↑ TOP

No errors were detected in phenylalanine torsion angle conventions.

11. Note: Aspartic acid torsion conventions OK Pass

↑ TOP

No errors were detected in aspartic acid torsion angle conventions.

12. Note: Glutamic acid torsion conventions OK Pass

↑ TOP

No errors were detected in glutamic acid torsion angle conventions.

13. Note: L or D conformation checked OK Pass

↑ TOP

All amino acids in the structure have the L conformation.

14. Note: Chain names are unique Pass

↑ TOP

All chain names assigned to polymer molecules are unique.

15. Note: Weights checked OK Pass

↑ TOP

All atomic occupancy factors ('weights') fall in the 0.0--1.0 range.

16. Note: No missing atoms detected Pass

↑ TOP

All expected atoms are present.

17. Note: All bond lengths OK Pass

↑ TOP

All bond lengths are in agreement with standard bond lengths using
a tolerance of 4 sigma (both standard values and sigma for amino acid
residues have been taken from Engh and Huber [REF])

18. Note: Normal bond length variability Pass

↑ TOP

Bond lengths were found to deviate normally from the standard bond
lengths (values for Protein residues were taken from Engh and Huber
[REF]).

Z-score for bond lengths: 0.931
RMS-deviation in bond distances: 0.018

19. Warning: Directionality in bond lengths Warning

↑ TOP

Comparison of bond distances with Engh and Huber [REF] standard
values shows a significant systematic deviation. The bonds in one
direction are systematically longer than in other directions.

If this is not an XRAY structure this effect is hard to explain.
Otherwise you will have seen symmetry problems earlier. Please
correct these and rerun this check to see the implications on the
cell axes.

20. Warning: Unusual bond angles Warning

↑ TOP

The bond angles listed in the table below were found to deviate
more than 4 sigma from standard bond angles (both standard values
and sigma have been taken from Engh and Huber [REF]). In the table
below for each strange angle the bond angle and the number of
standard deviations it differs from the Engh and Huber values is
given. Please note that only bond angles within protein residues
are taken into account: disulphide bridges and peptide bonds are
neglected.

15	HIS	(15)	- NE2	CD2	CG	110.624	4.1
19	ASN	(19)	- CA	CB	CG	106.602	-6.0
27	ASN	(27)	- CA	CB	CG	107.708	-4.9
34	PHE	(34)	- CA	CB	CG	103.926	-9.9
37	ASN	(37)	- CA	CB	CG	108.371	-4.2
38	PHE	(38)	- CA	CB	CG	109.071	-4.7
41	GLN	(41)	- CB	CG	CD	104.524	-4.8
44	ASN	(44)	- CA	CB	CG	108.383	-4.2
46	ASN	(46)	- CA	CB	CG	107.984	-4.6
52	ASP	(52)	- CA	CB	CG	108.427	-4.2
63	TRP	(63)	- N	CA	C	122.787	4.1
74	ASN	(74)	- CA	CB	CG	107.796	-4.8
93	ASN	(93)	- CA	CB	CG	108.043	-4.6
101	ASP	(101)	- CA	CB	CG	108.351	-4.2
103	ASN	(103)	- CA	CB	CG	108.442	-4.2
106	ASN	(106)	- CA	CB	CG	106.565	-6.0
113	ASN	(113)	- CA	CB	CG	107.488	-5.1

21. Note: Normal bond angle variability Pass

↑ TOP

Bond angles were found to deviate normally from the mean Engh and
Huber [REF] standard bond angles. The RMS Z-score given below is
expected to be around 1.0 for a normally restrained data set, and
this is indeed observed for very high resolution X-ray structures.
More common values are around 1.55

Z-score for bond angles: 1.276
RMS-deviation in bond angles: 2.142

22. Note: Side chain planarity OK Pass

↑ TOP

All of the side chains of residues that have a planar group are
planar within expected RMS deviations.

23. Note: Atoms connected to aromatic rings OK Pass

↑ TOP

```
All of the atoms that are connected to planar aromatic rings in side
chains of amino-acid residues are in the plane within expected RMS
deviations.
```

24. Note: PRO puckering amplitude OK Pass

↑ TOP

```
Puckering amplitudes for all PRO residues are within normal ranges.
```

25. Warning: Strange PRO puckering phases Warning

↑ TOP

```
The proline residues listed in the table below have a puckering phase
that is not expected to occur in protein structures. Puckering
parameters were calculated by the method of Cremer and Pople
[REF]. Normal PRO rings approximately show a so-called envelope
conformation with the C-gamma atom above the plane of the ring
(phi=+72 degrees), or a half-chair conformation with C-gamma below
and C-beta above the plane of the ring (phi=-90 degrees). If phi
deviates strongly from these values, this is indicative of a very
strange conformation for a PRO residue, and definitely requires a
manual check of the data.


  70 PRO  (70  ) - -121.4 half-chair C-delta/C-gamma (-126 degrees)
  79 PRO  (79  ) - -119.4 half-chair C-delta/C-gamma (-126 degrees)
```

26. Note: Torsion angles OK Pass

↑ TOP

```
All individual residues have normal overall torsion angle scores.
```

27. Warning: Backbone torsion angle evaluation shows unusual conformations Warning

↑ TOP

```
The residues listed in the table below have abnormal backbone torsion
angles.

Residues with ``forbidden'' phi-psi combinations are listed, as
well as residues with unusual omega angles (deviating by more than
3 sigma from the normal value). Please note that it is normal if
about 5 percent of the residues is listed here as having unusual
phi-psi combinations.

  19 ASN  (19  ) - Poor phi/psi
  21 ARG  (21  ) - Poor phi/psi
  37 ASN  (37  ) - Poor phi/psi
  38 PHE  (38  ) - Poor phi/psi
  57 GLN  (57  ) - Poor phi/psi
  62 TRP  (62  ) - omega poor
  77 ASN  (77  ) - Poor phi/psi
```

28. Note: Ramachandran Z-score OK Pass

↑ TOP

```
The score expressing how well the backbone conformations of all residues
are corresponding to the known allowed areas in the Ramachandran plot is
within expected ranges for well-refined structures.

 Ramachandran Z-score : 0.764
```

29. Warning: Omega angles too tightly restrained Warning

↑ TOP

```
The omega angles for trans-peptide bonds in a structure are
expected to give a gaussian distribution with the average around
+178 degrees and a standard deviation around 5.5 degrees. These
expected values were obtained from very accurately determined
structures.  Many protein structures are too tightly constrained.
This seems to be the case with the current structure, as the
observed standard deviation is below 4.0 degrees.

 Standard deviation of omega values : 3.847
```

30. Note: chi-1/chi-2 angle correlation Z-score OK Pass

↑ TOP

```
The score expressing how well the chi-1/chi-2 angles of all residues
are corresponding to the populated areas in the database is
within expected ranges for well-refined structures.

 chi-1/chi-2 correlation Z-score : -0.479
```

31. Note: Ramachandran plot Pass

↑ TOP

```
In this Ramachandran plot large crosses represent glycines and
small crosses represent the other residues. If too many small crosses
fall outside the boxed areas then the molecule is poorly refined (or
worse).

In the TeX file, a plot has been inserted here

 Chain without chain identifier
```

32. Note: Secondary structure Pass

↑ TOP

```
This is the secondary structure according to DSSP. Only helix (H), strand
(S), turn (T) and coil (blank) are shown. [REF]

 Secondary structure assignment
The DSSP executable was not found
/software/what/whattest/dssp/DSSP.EXE
WARNING. You don't have the DSSP program installed. Therefore
the emulator will be used. This emulator gives rather poor results,
but it prevents WHAT IF from crashing. See the writeup about this.
                    10        20        30        40        50        60
                     |         |         |         |         |         |
    1 -  60  KVFGRCELAAAMKRHGLDNYRGYSLGNWVCAAKFESNFNTQATNRNTDGSTDYGILQINS
    1 -  60     THHHHHHHHHHHT3 TTTTTTHHHHHHHHHHHTTT  T S T  T  SS   TST
                    70        80        90       100       110       120
                     |         |         |         |         |         |
   61 - 120  RWWCNDGRTPGSRNLCNIPCSALLSSDITASVNCAKKIVSDGNGMNAWVAWRNRCKGTDV
   61 - 120    H S T   T  T  TST33333T  HHHHHHHHHH33H3T3 333THH HHH3 T3T3


  121 - 129  QAWIRGCRL
  121 - 129  H333TTT
```

33. Error: Abnormally short interatomic distances Error

↑ TOP

The pairs of atoms listed in the table below have an unusually short distance.

The contact distances of all atom pairs have been checked. Two atoms are said to `bump' if they are closer than the sum of their Van der Waals radii minus 0.40 Angstrom. For hydrogen bonded pairs a tolerance of 0.55 Angstrom is used. The first number in the table tells you how much shorter that specific contact is than the acceptable limit. The second distance is the distance between the centers of the two atoms.

The last text-item on each line represents the status of the atom pair. The text `INTRA' means that the bump is between atoms that are explicitly listed in the PDB file. `INTER' means it is an inter-symmetry bump. If a line contains the text 'HB', the bump criterium was relaxed because there could be a hydrogen bond. If the text 'BF' is present, the sum of the B-factors of the atoms is higher than 80, which makes the appearance of the bump somewhat less severe because the atoms probably aren't there anyway.

Bumps between atoms for which the sum of their occupancies is lower than one are not reported. In any case, each bump is listed in only one direction.

109 VAL	(109) - CG2	--	110 ALA	(110) - N	0.279	2.821	INTRA BF
55 ILE	(55) - CG2	--	56 LEU	(56) - N	0.272	2.828	INTRA BF
13 LYS	(13) - CE	--	129 LEU	(129) - CB	0.257	2.943	INTRA BF
1 LYS	(1) - CE	--	3 PHE	(3) - CZ	0.255	2.945	INTRA BF
33 LYS	(33) - CD	--	38 PHE	(38) - CE2	0.254	2.946	INTRA BF
1 LYS	(1) - CE	--	3 PHE	(3) - CE2	0.227	2.973	INTRA BF
109 VAL	(109) - CG1	--	112 ARG	(112) - NH2	0.224	2.876	INTRA BF
8 LEU	(8) - CD2	--	38 PHE	(38) - CE1	0.222	2.978	INTRA BF
45 ARG	(45) - NH2	--	68 ARG	(68) - NH1	0.220	2.780	INTRA BF
111 TRP	(111) - CD1	--	115 CYS	(115) - CB	0.215	2.985	INTRA BF
58 ILE	(58) - CG2	--	63 TRP	(63) - CB	0.201	2.999	INTRA BF
6 CYS	(6) - CA	--	127 CYS	(127) - SG	0.174	3.226	INTRA
51 THR	(51) - CG2	--	52 ASP	(52) - N	0.163	2.937	INTRA BF
8 LEU	(8) - CD2	--	38 PHE	(38) - CD1	0.139	3.061	INTRA BF
64 CYS	(64) - C	--	80 CYS	(80) - SG	0.129	3.271	INTRA BF
99 VAL	(99) - CG1	--	100 SER	(100) - N	0.127	2.973	INTRA BF
45 ARG	(45) - CG	--	46 ASN	(46) - N	0.119	2.981	INTRA BF
40 THR	(40) - CG2	--	85 SER	(85) - O	0.094	2.706	INTRA BF
24 SER	(24) - N	--	27 ASN	(27) - ND2	0.090	2.910	INTRA BF
39 ASN	(39) - OD1	--	41 GLN	(41) - CG	0.065	2.735	INTRA BF
103 ASN	(103) - OD1	--	106 ASN	(106) - CB	0.060	2.740	INTRA BF
15 HIS	(15) - CE1	--	88 ILE	(88) - CG1	0.043	3.157	INTRA BF
30 CYS	(30) - C	--	115 CYS	(115) - SG	0.032	3.368	INTRA BF
120 VAL	(120) - CG2	--	121 GLN	(121) - N	0.029	3.071	INTRA BF
5 ARG	(5) - CD	--	123 TRP	(123) - O	0.028	2.772	INTRA BF
15 HIS	(15) - ND1	--	88 ILE	(88) - CD1	0.025	3.075	INTRA BF
27 ASN	(27) - CB	--	105 MET	(105) - SD	0.018	3.382	INTRA BF
121 GLN	(121) - O	--	124 ILE	(124) - CG1	0.016	2.784	INTRA BF
53 TYR	(53) - CE2	--	60 SER	(60) - OG	0.014	2.786	INTRA BF
23 TYR	(23) - CD1	--	105 MET	(105) - SD	0.005	3.395	INTRA BF

34. Warning: Abnormal packing environment for some residues Warning

↑ TOP

The residues listed in the table below have an unusual packing environment.

The packing environment of the residues is compared with the average packing environment for all residues of the same type in good PDB files. A low packing score can indicate one of several things: Poor packing, misthreading of the sequence through the density, crystal contacts, contacts with a co-factor, or the residue is part of the active site. It is not uncommon to see a few of these, but in any case this requires further inspection of the residue.

128 ARG	(128)	-8.43
68 ARG	(68)	-6.80
45 ARG	(45)	-6.39
14 ARG	(14)	-5.72
15 HIS	(15)	-5.62
125 ARG	(125)	-5.62
103 ASN	(103)	-5.38
21 ARG	(21)	-5.12
113 ASN	(113)	-5.04

35. Note: No series of residues with bad packing environment Pass

↑ TOP

There are no stretches of three or more residues each having a quality control score worse than -4.0.

36. Note: Structural average packing environment OK Pass

↑ TOP

The structural average quality control value is within normal ranges.

Average for range 1 - 129 : -1.316

37. Note: Backbone oxygen evaluation OK Pass

↑ TOP

All residues for which the local backbone conformation could be found in the WHAT IF database have a normal backbone oxygen position.

38. Note: Rotamers checked OK Pass

↑ TOP

None of the residues that have a normal backbone environment have abnormal rotamers.

39. Warning: Unusual backbone conformations Warning

↑ TOP

For the residues listed in the table below, the backbone formed by itself and two neighboring residues on either side is in a conformation that is not seen very often in the database of solved protein structures. The number given in the table is the number of similar backbone conformations in the database with the same amino acid in the center.

For this check, backbone conformations are compared with database structures using C-alpha superpositions with some restraints on the backbone oxygen positions.

A residue mentioned in the table can be part of a strange loop, or there might be something wrong with it or its directly surrounding residues. There are a few of these in every protein, but in any case it is worth looking at!

3 PHE	(3)	0
4 GLY	(4)	0
5 ARG	(5)	0
6 CYS	(6)	0
7 GLU	(7)	0
8 LEU	(8)	0
9 ALA	(9)	0
10 ALA	(10)	0
11 ALA	(11)	0
12 MET	(12)	0

```
  13 LYS  (13  )    0
  14 ARG  (14  )    0
  15 HIS  (15  )    0
  16 GLY  (16  )    0
  17 LEU  (17  )    0
  18 ASP  (18  )    0
  19 ASN  (19  )    0
  20 TYR  (20  )    0
  21 ARG  (21  )    0
  22 GLY  (22  )    0
  23 TYR  (23  )    0
  24 SER  (24  )    0
  25 LEU  (25  )    0
  26 GLY  (26  )    0
  27 ASN  (27  )    0
And so on for a total of   125 lines
```

40. Error: Backbone conformation Z-score very low **Error**

↑TOP

```
A comparison of the backbone conformation with database proteins
shows that the backbone fold in this structure is very unusual.

 Backbone conformation Z-score : -30.011
```

41. Note: Symmetry related water molecules check not performed Pass

↑TOP

```
Since there is no symmetry, the position check for symmetry related
water molecules can not be performed
```

42. Error: Average B-factor error **Error**

↑TOP

```
The average B-factor for all buried protein atoms normally lies between
10--20. Values around 3--5 are expected for X-ray studies performed
at liquid nitrogen temperature.

Because of the extreme value for the average B-factor, no further analysis
of the B-factors is performed.

Average B-factor for buried atoms : 63.122
```

43. Error: HIS, ASN, GLN side chain flips **Error**

↑TOP

```
Listed here are Histidine, Asparagine or Glutamine residues for which
the assignment or orientation determined from hydrogen bonding analysis
are different from the assignment given in the input. Either they could
form energetically more favorable hydrogen bonds if the terminal group
was rotated by 180 degrees, or there is no assignment in the input
file (atom type 'A') but an assignment could be made.

  46 ASN  (46  )
 121 GLN  (121 )
```

44. Note: Histidine type assignments Pass

↑TOP

```
For all complete HIS residues in the structure a tentative
assignment to HIS-D (protonated on ND1), HIS-E (protonated on NE2),
or HIS-H (protonated on both ND1 and NE2, positively charged) is
made based on the hydrogen bond network.  A second assignment is
made based on which of the Engh and Huber [REF] histidine
geometries fits best to the structure.

In the table below all normal histidine residues are listed. The
assignment based on the geometry of the residue is listed first,
together with the RMS Z-score for the fit to the Engh and Huber
parameters.  For all residues where the H-bond assignment is
different, the assignment is listed in the last columns, together
with its RMS Z-score to the Engh and Huber parameters.

As always, the RMS Z-scores should be close to 1.0 if the residues
were restrained to the Engh and Huber parameters during refinement.

Please note that because the differences between the geometries of
the different types are small it is possible that the geometric
assignment given here does not correspond to the type used in
refinement.  This is especially true if the Z-scores are much higher
than 1.0.

If the two assignments differ, or the ``geometry'' Z-score is high,
it is advisable to verify the hydrogen bond assignment, check the
HIS type used during the refinement and possibly adjust it.

  15 HIS  (15  ) - HIS-D   0.30 HIS-H   1.66
```

45. Warning: Buried unsatisfied hydrogen bond donors Warning

↑TOP

```
The buried hydrogen bond donors listed in the table below have a
hydrogen atom that is not involved in a hydrogen bond in the
optimized hydrogen bond network.

Hydrogen bond donors that are buried inside the protein normally
use all of their hydrogens to form hydrogen bonds within the
protein. If there are any non hydrogen bonded buried hydrogen bond
donors in the structure they will be listed here. In very good
structures the number of listed atoms will tend to zero.

  18 ASP  (18  ) - N
  51 THR  (51  ) - N
  51 THR  (51  ) - OG1
  55 ILE  (55  ) - N
  62 TRP  (62  ) - N
  66 ASP  (66  ) - N
 111 TRP  (111 ) - N
```

46. Note: Buried hydrogen bond acceptors OK Pass

↑TOP

```
All buried polar side-chain hydrogen bond acceptors are involved in a
hydrogen bond in the optimized hydrogen bond network.
```

47. Note: Overall summary report Pass

↑TOP

```
This is an attempt to create an overall summary of the quality of the
structure. We do not recommend anyone to look at these numbers, please
look at the complete report instead.

 Structure Z-scores, positive is better than average:
  1st generation packing quality :  -2.040
  Ramachandran plot appearance   :   0.764
  chi-1/chi-2 rotamer normality  :  -0.479
  Backbone conformation          : -30.011 (bad)

 RMS Z-scores, should be close to 1.0:
  Bond lengths                   :   0.931
  Bond angles                    :   1.276
  Omega angle restraints         :   0.699 (tight)
```

```
  Side chain planarity          :   0.269 (tight)

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```

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